

2026 Practice Set – Round 3

TOSS-UP

1) MATH *Short Answer* What is the value of $3x^2 - 2x + 1$ when $x = 2$?

ANSWER: 9

BONUS

1) MATH *Short Answer* If $f(x) = x^2 + 1$, what is $f(x + 1) - f(x)$ when $x = 5$?

ANSWER: 11

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#### TOSS-UP

2) BIOLOGY *Short Answer* What organelle is the site of aerobic respiration in eukaryotic cells?

ANSWER: MITOCHONDRION (ACCEPT: MITOCHONDRIA)

#### BONUS

2) BIOLOGY *Multiple Choice* Which of the following is a prokaryote?

- W) Yeast
- X) Amoeba
- Y) E. coli
- Z) Paramecium

ANSWER: Y) E. COLI

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3) MATH *Multiple Choice* Which of the following is equal to  $-2^4$ ?

W)  $-16$

X)  $16$

Y)  $-8$

Z)  $8$

ANSWER: W)  $-16$

BONUS

3) MATH *Short Answer* Solve for x:  $2^{x+1} = 32$ .

ANSWER: 4

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TOSS-UP

4) PHYSICS *Short Answer* What is the SI unit of force?

ANSWER: NEWTON

BONUS

4) MATH *Short Answer* What is the value of $\log_2 64 - \log_2 8$?

ANSWER: 3

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TOSS-UP

5) MATH *Short Answer* What is the value of $(2x + 1)^2 - 2x^2$ when $x = 3$?

ANSWER: 31

BONUS

5) MATH *Short Answer* What is the area, in square units, of the triangle with vertices $(0, 0)$, $(6, 0)$, and $(0, 4)$?

ANSWER: 12